DAY-10(29.07.2026)

▼ HOW DO IP ADDRESS WORK?

- governs how data is packaged, addressed, transmitted, routed, received

UNIQUE IDENTIFICATION:

Every device connected to network → assigned with IP address →to send or receive data →to identify the device

COMMUNICATION PROTOCOL:

IP → how data is transmitted b/w device → as packets → routers , switches

DATA ROUTING:

How the data is sent :

1. data → packets
2. packets → has IP of its destination
3. routers → best route → based on the IP
4. routers→ communicates → to update routing → maintains most efficient path of transmission

LAN AND WAN COMMUNICATION:

LAN → IP address(statically/dynamically) →DHCP →communicates with same LAN using private IP

WAN →data travels through multiple routers over internet

NETWORK ADDRESS TRNSLATION:

multiple devices → in private IP → allows to share single public IP → to access the internet

translates private IP → public IP → allowing the device on private network to access with internet

▼ IP ADDRESS SECURITY THREATS:

1. **IP spoofing:**
2. **DDoS attacks:**
3. **Man-in-the-Middle:**
4. **Port Scanning:**

SUBNETTING:

it is a process of dividing large IP network into smaller logical networks called subnet

1. Network portion → identifies the network to which device belongs
2. Host portion → specific device within network

CLASSES OF IPv4 BASED ON HOW MANY BITS ARE USED FOR NETWORK ID AND HOST ID

1. class A →8bit (network), 24bit(host)
2. class B →24bit(network), 24 bit(host)
3. class c →24bit(network), 8bit(host)

SUBNET MASK:

32-bit number used in IP addressing to separate the network portion of an IP address from the host portion.